Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/12/2010
PI Name:	Shoukas, Artin A. Ph.D.		
Project Title:	Radiation, Endothelial Cell Senescence, Accelerated Aging, and Atherosclerosis		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21205-2109	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-01 Human Health in Space
Start Date:	03/01/2008	**End Date:**	02/28/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Berkowitz, Dan (The Johns Hopkins University) Nyhan, Daniel (The Johns Hopkins University) Flavahan, Nicholas (Johns Hopkins Medical School) Romer, Lewis (Johns Hopkins Medical School) DeWeese, Theodore (Johns Hopkins Medical School)		
Grant/Contract No.:	NCC 9-58-CA01301		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Exposure to ionizing radiation represents one of the most significant health risks to astronauts during both short, but particularly long, duration spaceflight. Vascular endothelial cells (ECs), the most radiosensitive vascular cell type, remain the major regulating interface between the blood and vessel wall. There, they act as a paracrine organ releasing signaling molecules that regulate vessels tone, permeability and cell and platelet adhesion. Dysfunction of the endothelium may result from both an increase in endothelial cell damage (oxidative stress with nitroso-redox imbalance), an impairment of the endothelial repair mechanisms (decrease in the number and function of endothelial progenitor cells) or an acceleration of endothelial cell senescence. Radiation is known to induce cellular oxidative stress, as well as suppress bone marrow proliferation. Data from our lab suggests that gamma and heavy iron ion radiation results in the activation of the ROS producing enzyme xanthine oxidase (XO). Activation of enzymes such as arginase results in a decrease in endothelial protective nitric oxide (NO) with altered nitroso-redox balance and endothelial damage. We hypothesize that radiation will result in activation of ROS producing enzymes that will result in increased oxidative stress, nitroso-redox imbalance and endothelial cell damage. We also hypothesize that radiation will decrease endothelial repair mechanism by reducing bone marrow derived endothelial progenitor cells (EPCs). Additionally, we hypothesize that radiation will accelerate endothelial cell senescence by decreasing telomerase activity and increasing the rate of telomere attrition. This will ultimately result in significant endothelial dysfunction, mimic the aging phenotype and accelerate the development of atherosclerosis. We will test the countermeasure hypothesis that the XO inhibitor (oxypurinol) will reduce radiation related endothelial dysfunction, that arginase inhibition (which reciprocally regulates nitric oxide synthase) will restore nitroso-redox balance and promote endothelial cell protection. We will also test the hypothesis that statins (endothelial cell protectants through a number of pleotropic effects independent of their lipid lowering effects) will prevent oxidative stress, improve nitroso-redox imbalance, enhance endothelial progenitor cell production, and prevent telomere decay. In this way, the adverse effects of radiation on the endothelial cell will be attenuated. We have demonstrated that radiation-induced endothelial dysfunction is largely due to the activation of xanthine oxidase in rat aorta as reactive oxygen species derived from xanthine oxidase contribute to impaired endothelial nitric oxide production and attenuated endothelial-dependent vasorelaxation. In the future, we hope to investigate further mechanisms contributing to vascular radiation injury through the use of knockout mouse models. Therefore, during the past year we have begun to characterize the vascular response to radiation of mice. In addition, we are interested in intrinsic endothelial defense mechanisms, and whether these mechanisms are dysregulated by radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The current work being performed has the potential to protect humans undergoing either diagnostic radiation procedures or radiation treatment for cancerous tumors. Development of counter measures for the protection of astronauts may be applicable to prevent morbidities seen in patients undergoing radiation therapy for cancerous tumors.
Task Progress:	Summary for Years 1 and Year 2: The DHE fluorescence rate in the aorta segments of rats exposed to 1 Gy was significantly greater than that of control rats (8.45e-5±0.733e-5 sec-1 versus 4.84e-5±1.29e-5 sec-1, p<0.05). Furthermore, 30 minutes of Oxp incubation significantly attenuated the DHE response in irradiated aorta. Following 10-5M ACh administration, aortas of 0Gy rats demonstrated a 25% fluorescent rate increase from baseline slope (125±8.42%), while aortas of the 1Gy-irradiated rats have a negligible response (97.0±6.22%). Dietary XO inhibition elevated this response significantly, compared to untreated, 1Gy irradiated rats (159±19.4%, P<0.05). The SNP response between groups is similar, implying that DAF-FM loading is consistent and sufficient. In the study of chronic XO inhibition, rats exposed to 100cGy radiation possessed a significantly elevated Pulse Wave Velocity (PWV) (4.45±0.244 m/sec, n=8) from 0cGy-exposed rats (n=10, P<0.01) and 100cGy Oxp-diet rats (n=6, P<0.05). The PWV measurements in 0cGy and 100cGy+Oxp rats are not statistically different. The aorta of 5Gy irradiated rats demonstrate a significantly elevated XO activity from controls (P<0.05, unpaired Student t test). Remarkably, this activation is abolished by Oxp dietary inhibition ,endothelium removal (P<0.05, paired Student t test), and allopurinol treatement (P<0.001, paired Student t test). Total XO+XDH activity was not significantly affected by radiation or Oxp diet. As a result, the aortic XO:XDH activity ratio was increased by 5Gy-radiation, whereas Oxp treatment significantly reduced the ratio (P<0.001, unpaired Student t test). Aortas of the irradiated mice groups demonstrated statistically significant increase in PE-induced constriction tensions compared to controls. Trends seen in KCl-induced constriction tensions were not statistically significant. Tie2 is an endothelial-specific receptor protein. Thus, cells that emit GFP signal would identify the endothelial cell phenotype. We observed robust GFP signal in the cell outgrowth, suggesting that the cellular outgrowth is largely composed of endothelial cell (data not shown). We are beginning to investigate the role of thioredoxin in normal Nrf2 signaling.
Bibliography Type:	Description: (Last Updated: 05/05/2011)
Articles in Peer-reviewed Journals	Soucy KG, Attarzadeh DO, Ramachandran R, Soucy PA, Romer LH, Shoukas AA, Berkowitz DE. "Single exposure to radiation produces early anti-angiogenic effects in mouse aorta." Radiat Environ Biophys 2010 Mar; in press. http://dx.doi.org/10.1007/s00411-010-0287-z , Mar-2010
Articles in Peer-reviewed Journals	Soucy KG, Lim HK, Attarzadeh DO, Santhanam L, Kim JH, Bhunia AK, Sevinc B, Ryoo S, Vazquez ME, Nyhan D, Shoukas AA, Berkowitz DE. "Dietary inhibition of xanthine oxidase attenuates radiation-induced endothelial dysfunction in rat aorta." J Appl Physiol. Epub 2010 Feb 18. PubMed PMID: 20167676. , Feb-2010
Awards	Soucy K. "Johns Hopkins University Graduate Student Association Travel Award, May 2009." May-2009
Awards	Soucy K. "NASA Student Travel Award, Cologne, Germany, July 2009." Jul-2009

Papers from Meeting Proceedings	Attarzadeh D, Soucy KG, Ramachandran R, Soucy PA, Sevinc B, Romer LH, Shoukas AA, Berkowitz DE. "Attenuation of angiogenesis in mouse aorta following radiation exposure: a meta-analysis." 11th Annual Anesthesiology and Critical Care Medicine Research Day, Baltimore, MD, December 14, 2009. 11th Annual Anesthesiology and Critical Care Medicine Research Day, Baltimore, MD, December 14, 2009. , Dec-2009
Papers from Meeting Proceedings	Kuo M, Soucy KG, Shoukas AA, Berkowitz DE. "Low-dose iron radiation impairs vessel function." NASA Human Research Program Investigators's Workshop, Houston, TX, February 3-5, 2010. NASA Human Research Program Investigators's Workshop, Houston, TX, February 3-5, 2010. , Feb-2010
Papers from Meeting Proceedings	Soucy KG, Bhunia AK, Attarzadeh D, Sevinc B, Chang F, Romer LH, Nyhan D, Shoukas AA, Berkowitz DE. "High-dose radiation produces anti-angiogenic effects potentially through impaired Nrf2-antioxidant defenses." 5th International Workshop on Space Radiation Research and 20th Annual NASA Space Radiation Investigators' Workshop, Cologne, Germany, July 6-10, 2009. 5th International Workshop on Space Radiation Research and 20th Annual NASA Space Radiation Investigators' Workshop, Cologne, Germany, July 6-10, 2009. , Jul-2009
Papers from Meeting Proceedings	Soucy KG, Bhunia AK, Chang F, Attarzadeh D, Romer LH, Shoukas AA, Berkowitz DE. "Radiation-induced endothelial dysfunction: Investigating the balance of injury and repair." 2008 Biomedical Engineering Society Annual Fall Meeting, St. Louis, MO, October 2-4, 2008. 2008 Biomedical Engineering Society Annual Fall Meeting, St. Louis, MO, October 2-4, 2008. , Oct-2008
Patents	Patented. Patented, February 2010. Feb-2010 Cole RN, Berkowitz DE, Santhanam L, Shoukas AA. "Selective fluorescent labeling of S-nitrosothiols (S-FLOS): a novel method for studying S-nitrosylation."